E-JEPA-TTC: Dense Event-Token Joint-Embedding Predictive Pretraining for Time-to-Collision Estimation

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Local reproducible report generated from the `e-jepa-ttc` repository
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Abstract

Event cameras provide asynchronous measurements with high temporal resolution, making them attractive for time-to-collision (TTC) estimation in fast driving scenarios. This paper presents E-JEPA-TTC, a self-supervised event-camera representation-learning system for TTC estimation on a local EvTTC full-starter protocol. The system replaces an initial TinyCNN baseline with tokenized event transformers and then moves toward a V-JEPA-like objective: dense temporal token prediction, spatio-temporal event tubelet masking, causal ego-motion/action conditioning, an exponential-moving-average target encoder, and a transformer predictor over dense latent tokens. The best local all-window model, an event-tubelet transformer pretrained with tubelet masking and a dense transformer JEPA predictor, reaches 0.231 ± 0.018 s validation MAE and 0.312 ± 0.044 s diagnostic CPLA-high MAE over three fine-tuning seeds. Its mean absolute relative error is $8.19 \pm 0.53\%$ on validation and $6.42 \pm 0.45\%$ on CPLA-high. Low-label experiments show the strongest self-supervised signal: at 10% labels, token JEPA improves diagnostic CPLA-high MAE by 65.4% over the matching scratch token transformer.

The result is the strongest local all-window result in this repository, but it is not an official EvTTC state-of-the-art claim. The official EvTTC benchmark uses a bbox/ROI-assisted Table V protocol over CCRs1, CCRs2, CCRm, and slider sequences. An automated coverage gate implemented in the repository finds only 3/8 complete real-world official rows (37.5%) and 3/10 complete Table V rows including slider (30.0%). Detection-assisted ROI probers and predicted latent rollout probers were implemented to align with recent SkyJEPA and world-model practice, but their current flat-head forms are diagnostic rather than competitive. The contribution is therefore a rigorous, reproducible engineering path toward dense event-token JEPA for TTC, including anti-leakage discipline, negative ablations, and explicit claim boundaries.

Keywords: event cameras; time-to-collision; time-to-contact; JEPA; self-supervised learning; V-JEPA; world models; dense tokens; autonomous driving.

1 Introduction

Time-to-collision (TTC), also called time-to-contact in some geometric settings, is a direct measure of collision risk. For a target at range $d(t)$ with closing speed $v_c(t) > 0$, a first-order TTC estimate is

$$\tau(t) = \frac{d(t)}{v_c(t)}. \tag{1}$$

In real driving scenes, direct access to $d(t)$ and $v_c(t)$ may require calibrated depth, ego-motion, object localization, and robust temporal association. Event cameras offer a complementary sensor modality: instead of frame intensity, they report asynchronous brightness-change events at high temporal resolution. A single event stream can be represented as

$$\mathcal{E} = \{e_i\}_{i=1}^{N}, \qquad e_i = (x_i, y_i, t_i, p_i), \quad p_i \in \{-1, +1\}. \tag{2}$$

This representation is naturally suited to fast motion and temporal geometry, but it also creates a learning problem: the useful signal is sparse, asynchronous, and strongly dependent on motion.

The hypothesis explored here is that a predictive latent model can learn event dynamics without TTC labels and that those representations can improve TTC regression after supervised fine-tuning. The work follows the Joint-Embedding Predictive Architecture (JEPA) principle: predict target representations in latent space rather than reconstructing raw pixels or event tensors. This choice is attractive for event streams because it encourages predictive abstractions while avoiding the burden of reconstructing sparse high-frequency event noise.

The project described in this paper is deliberately conservative about claims. It contains strong local results on an EvTTC full-starter subset, but it does not reproduce the official EvTTC Table V benchmark. The paper therefore separates: (i) local all-window event-model performance, (ii) local detection-assisted bbox/ROI diagnostics, and (iii) official SOTA eligibility. Only the first two are supported by the current workspace.

1.1 Contributions

This paper makes the following concrete contributions:

1. A reproducible dense event-token JEPA pipeline for TTC estimation, including event voxelization, token-transformer backbones, event-tubelet tokenization, EMA target encoders, dense future-token prediction, and supervised log-TTC fine-tuning.

2. A V-JEPA-like event objective with spatio-temporal tubelet masking and a transformer dense predictor conditioned on causal ego-motion/action features.

3. A strict anti-leakage protocol: no TTC labels in self-supervised pretraining, sequence-level splits, context-only navigation/action features, train-only normalization, and checkpoint-only evaluation.

4. Extensive local experiments over scratch baselines, token JEPA, navigation-conditioned JEPA, event-tubelet JEPA, deep/self-supervision variants, all-token context losses, ROI latent probers, and predicted rollout probers.

5. An automated official EvTTC coverage gate that prevents unsupported SOTA claims until the required bbox/ROI benchmark assets and baseline wrappers are present.

2 Related Work

Joint-embedding predictive learning. I-JEPA introduced non-generative representation prediction for images: representations of visible context regions are trained to predict target representations of masked regions. V-JEPA extended this idea to video through masked spatio-temporal latent prediction. V-JEPA 2 and V-JEPA 2.1 further emphasize world-model behavior, dense features, temporal consistency, and scaling. The present work adapts these ideas to event-camera TTC by replacing image/video patches with event voxel/tubelet tokens and by predicting future event-window embeddings at multiple horizons.

Action-conditioned world models. LeWorldModel and related latent world-model approaches motivate conditioning future latent prediction on causal action or ego-motion. SkyJEPA motivates a two-stage recipe: learn action-conditioned latent dynamics, freeze the latent model, and train a lightweight physics-inspired prober. In E-JEPA-TTC, the driving analogue of action is the causal integrated-navigation vector and event-derived context-motion summary. The repo therefore implements action-conditioned JEPA prediction, frozen latent TTC probers, and predicted-rollout ROI probers.

Event-camera TTC. EvTTC is the core dataset target in this project. It provides event-camera streams, TTC labels, and benchmark comparisons against methods including STRTTC, CMax, ETTCM, FAITH, AEB-Tracker, and Image FoE. However, the published EvTTC benchmark is not simply an all-window event-only regression task. Its Table V comparison is bbox/ROI-assisted and evaluates specific CCRs/CCRm/slider sequences. This distinction is central to the claim boundary in this paper.

3 Dataset and Protocol

3.1 Local Full-Starter Protocol

The local full-starter protocol uses recovered EvTTC sequences under `datasets/evttc`. The main cache is the full-starter navigation voxel cache under `artifacts/features`. It contains 3972 windows with shape $[3972, 21, 90, 160]$:

- 10 event channels from 5 temporal bins and 2 polarities;
- 2 metadata channels;
- 9 causal integrated-navigation channels.

The development multi-validation split was introduced after CPLA-high had been opened. It keeps `CCRs-side-high` as `validation_car`, moves `CPLA-medium` into `validation_pedestrian`, and reserves `CPLA-high` only for frozen diagnostic checks. This split is used for architecture selection and negative ablations.

3.2 Official EvTTC Coverage Gate

The official EvTTC Table V benchmark reports CCRs1, CCRs2, CCRm, and slider rows. The local workspace has CCRs1 plus side/CPLA starter sequences, but not the full CCRs2, CCRm, and slider set with bbox/ROI assets. This is enforced by a repository command:

```
$env:PYTHONPATH='src'
.\.venv\Scripts\e-jepa-ttc.exe data official-coverage
  --root datasets\evttc
  --output artifacts\metrics\evttc_official_table_v_coverage.
    json
```

The latest local result is:

$$C_{\text{real}} = \frac{3}{8} = 37.5\%, \qquad C_{\text{TableV}} = \frac{3}{10} = 30.0\%. \tag{3}$$

Missing real-world rows are CCRs2 low/medium/high and CCRm low/medium. Missing slider rows are `Slider-750` and `Slider-1000`. Therefore `official_sota_claim_allowed=false`.

4 Event Representation

4.1 Voxelization

For each anchor time t, the model consumes a causal context window $[t - \Delta_c, t]$. The event stream in the window is discretized into $B = 5$ temporal bins, two polarities,

Table 1: Local full-starter sequence-level split. CPLA-high is diagnostic because it had been inspected in previous branches before the final paper.

Split	Sequences	Windows
Train	`CCRs-1-low-100-overlap-100`, `CCRs-1-medium-100-overlap-100`, `CCRs-1-high-100-overlap-100`, `CCRs-side-low`, `CCRs-side-medium`, `CPLA-low`, `CPLA-medium`	3019
Validation	`CCRs-side-high`	475
Diagnostic test	`CPLA-high`	478

and a spatial grid of height $H = 90$ and width $W = 160$. Let π_x and π_y map sensor coordinates into the grid, and let

$$b_i = \left\lfloor B \frac{t_i - (t - \Delta_c)}{\Delta_c + \epsilon} \right\rfloor \in \{0, \dots, B-1\}. \tag{4}$$

The raw voxel tensor is computed by accumulating causal event indicators:

$$V_{b,p,y,x} = \sum_{i: e_i \in [t-\Delta_c, t]} I_i^b I_i^p I_i^y I_i^x, \tag{5}$$
$$I_i^b = \mathbf{1}[b_i = b], \quad I_i^p = \mathbf{1}[p_i = p],$$
$$I_i^y = \mathbf{1}[\pi_y(y_i) = y], \quad I_i^x = \mathbf{1}[\pi_x(x_i) = x].$$

The event channels are optionally concatenated with metadata channels and causal navigation channels:

$$X_t = \mathrm{concat}(V_t, M_t, N_t) \in \mathbb{R}^{C \times H \times W}. \tag{6}$$

For the navigation cache, $C = 21$.

4.2 Causal Navigation and Action Features

The navigation vector uses only samples in the current context window:

$$n_t = [\|v_t\|_2,\ v_{x,t}, v_{y,t}, v_{z,t},\ a_{x,t}, a_{y,t}, a_{z,t},\ \dot{\psi}_t,\ q_t]^\top, \tag{7}$$

where v_t is the most recent velocity vector, a_t is a finite-difference acceleration estimate, $\dot{\psi}_t$ is yaw-rate, and q_t is a validity flag. The predictor also receives an event-motion summary m_t containing causal event mass, temporal mass slope, centroid shift, and polarity balance. The action-conditioning vector is

$$a_t = \mathrm{LN}\left(W_a[m_t; n_t] + b_a\right), \tag{8}$$

with normalization statistics estimated from train context windows only.

5 Model

5.1 Event-Tubelet Encoder

The final backbone treats event channels as a polarity-by-time tensor rather than a flat channel stack. Let $X_t^{\mathrm{ev}} \in \mathbb{R}^{2 \times B \times H \times W}$ be the event component. A 3D tubelet embedding partitions it into local blocks

$$u_j = X_t^{\mathrm{ev}}[:,\ b_j : b_j + \delta_b,\ y_j : y_j + \delta_y,\ x_j : x_j + \delta_x], \tag{9}$$

and maps each tubelet to a token:

$$z_j^{(0)} = W_{\mathrm{tube}} \, \mathrm{vec}(u_j) + E_j^{\mathrm{pos}} + E_j^{\mathrm{aux}}. \tag{10}$$

E_j^{aux} injects spatially aligned metadata/navigation context. The transformer encoder then computes

$$Z_t = f_\theta(X_t) = \mathrm{Transformer}_\theta\left(\{z_j^{(0)}\}_{j=1}^P\right) \in \mathbb{R}^{P \times d}. \tag{11}$$

The supervised TTC head predicts log-TTC from a token summary:

$$\hat{y}_t = h_\phi\left(\mathrm{pool}(Z_t)\right), \qquad \hat{\tau}_t = \exp(\hat{y}_t) - \epsilon. \tag{12}$$

5.2 Dense Temporal JEPA

Self-supervised pretraining uses an online encoder f_θ, an EMA target encoder $f_{\bar{\theta}}$, and a predictor g_ψ. Given context window X_t and future target windows $X_{t+\delta}$ for horizons

$$\mathcal{H} = \{20, 60, 100, 240, 500\} \text{ ms}, \tag{13}$$

the target tokens are stop-gradient EMA outputs:

$$Y_{t,\delta} = \mathrm{sg}\left(f_{\bar{\theta}}(X_{t+\delta})\right). \tag{14}$$

The online context tokens are

$$Z_t = f_\theta(\tilde{X}_t), \tag{15}$$

where $\tilde{X}_t$ is a masked context view. The dense predictor estimates future token representations:

$$\hat{Y}_{t,\delta} = g_\psi(Z_t, a_t, e_\delta), \tag{16}$$

where e_δ is a learned horizon embedding. The dense JEPA loss is

$$\mathcal{L}_{\mathrm{JEPA}} = \frac{1}{|\mathcal{B}|\,|\mathcal{H}|\,P} \sum_{t \in \mathcal{B}} \sum_{\delta \in \mathcal{H}} \sum_{j=1}^{P} \left\| \hat{Y}_{t,\delta,j} - Y_{t,\delta,j} \right\|_2^2. \tag{17}$$

The target encoder is updated by EMA:

$$\bar{\theta} \leftarrow \mu \bar{\theta} + (1 - \mu)\theta, \qquad \mu = 0.99. \tag{18}$$

5.3 Tubelet Masking

The final pretraining variant masks spatio-temporal event tubelets while preserving metadata and navigation. For a mask set $\mathcal{M}$ sampled over event tubelets,

$$\tilde{X}^{\text{ev}}_{t,j} = \begin{cases} 0, & j \in \mathcal{M}, \\ X^{\text{ev}}_{t,j}, & j \notin \mathcal{M}, \end{cases} \qquad \frac{|\mathcal{M}|}{P_{\text{ev}}} \approx 0.45. \tag{19}$$

This matches the event structure better than purely spatial masking: the model must infer future latent dynamics from partially removed temporal event evidence, while still receiving causal ego-motion context.

5.4 Regularization and Negative Variants

Several SOTA-inspired variants were implemented:

- deep supervision on intermediate transformer layers;
- layer-aware prediction using predictor layer embeddings;
- larger token transformer backbones;
- all-token context loss;
- VISReg-style variance/sketching hooks;
- predicted-rollout feature modes.

The optional all-token context loss has the form

$$\mathcal{L}_{\text{ctx}} = \frac{1}{|\mathcal{B}|P} \sum_{t\in\mathcal{B}} \sum_{j=1}^{P} \|q_\psi(Z_t, a_t)_j - \text{sg}(f_{\bar{\theta}}(X_t))_j\|_2^2, \tag{20}$$

and the total objective is

$$\mathcal{L} = \mathcal{L}_{\text{JEPA}} + \lambda_{\text{ctx}}\mathcal{L}_{\text{ctx}} + \lambda_{\text{reg}}\mathcal{L}_{\text{reg}}. \tag{21}$$

On the current local data, all-token context loss and deep supervision were negative or neutral in the final selection protocol.

6 Supervised Fine-Tuning and Metrics

After self-supervised pretraining, the TTC head is trained on labels from the training split. The model predicts log-TTC:

$$y_t = \log(\tau_t + \epsilon), \qquad \hat{y}_t = h_\phi(f_\theta(X_t)). \tag{22}$$

The supervised loss is

$$\mathcal{L}_{\text{sup}} = \frac{1}{|\mathcal{B}|} \sum_{t\in\mathcal{B}} \ell(\hat{y}_t, y_t), \tag{23}$$

where ℓ is a robust regression loss in log-TTC space in the implementation family. Evaluation reports TTC-space metrics:

$$\text{MAE} = \frac{1}{N} \sum_{i=1}^{N} |\hat{\tau}_i - \tau_i|, \tag{24}$$

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^{N} (\hat{\tau}_i - \tau_i)^2}, \tag{25}$$

$$\text{MRE} = 100\frac{1}{N} \sum_{i=1}^{N} \frac{|\hat{\tau}_i - \tau_i|}{\max(\tau_i, \epsilon)}. \tag{26}$$

When users ask for a percentage-style interpretation, $100-\text{MRE}$ is an intuitive but informal accuracy-like number. It is not the benchmark metric. For the best CPLA-high diagnostic result, $\text{MRE} = 6.42\%$, which corresponds to approximately 93.58% in this informal reading.

7 Detection-Assisted ROI Probers

The repository also implements bbox/ROI diagnostics. These are closer to the official CMax/STRTTC input assumption, but they are frame-label-only and not comparable to all-window metrics. For a bounding box with width w_t and height h_t, a simple area-scale geometry reference uses

$$s_t = \sqrt{w_t h_t}. \tag{27}$$

If image-plane scale grows approximately as inverse depth, then TTC can be estimated from scale expansion:

$$\hat{\tau}_t \approx \frac{s_t}{\dot{s}_t + \epsilon}, \qquad \dot{s}_t \approx \frac{s_t - s_{t-\Delta}}{\Delta}. \tag{28}$$

The ROI event ridge and ROI latent prober use only events in $[t-\Delta_c, t]$ and the current object box. The prober form is

$$\hat{\tau}_t = r_\omega\left(\rho_{\text{ROI}}(E_{[t-\Delta_c,t]}, B_t), \text{pool}(f_\theta(X_t)), \hat{\tau}_t^{\text{ridge}}\right), \tag{29}$$

where B_t is the current box, ρ_{ROI} are causal ROI event features, and $\hat{\tau}_t^{\text{ridge}}$ is a train-only ridge prior.

The rollout prober uses frozen predicted future tokens:

$$\hat{Z}_{t+\delta} = g_\psi(f_\theta(X_t), a_t, e_\delta), \qquad \delta \in \mathcal{H}, \tag{30}$$

and trains a lightweight head over $\{\hat{Z}_{t+\delta}\}_{\delta\in\mathcal{H}}$. The current flat-summary version is a negative result, indicating that future work should use a constrained horizon-structured kinematic TTC head.

8 Anti-Leakage Discipline

The project explicitly separates self-supervised pretraining, validation selection, and diagnostic testing. The main controls are:

1. splits are sequence-level;
2. JEPA pretraining uses no TTC labels;
3. future event windows are self-supervised targets only;
4. target pairs do not cross sequence boundaries;
5. target pairs do not cross split boundaries;
6. navigation/action features are extracted from the current context only;
7. action normalization is fit on train context windows only;
8. ROI event features use $[t - \Delta_c, t]$ only;
9. no future boxes, future events, or future navigation enter ROI probers;
10. checkpoint-only evaluators reload saved checkpoints and do not retrain.

This discipline is why CPLA-high is treated as diagnostic: it had been inspected in earlier branches, so more tuning against it would be data snooping.

9 Experiments

9.1 Main All-Window Results

The strongest all-window model reaches:

$$\text{MAE}_{\text{val}} = 0.231477844 \pm 0.017632455 \text{ s}, \tag{31}$$
$$\text{MRE}_{\text{val}} = 8.192429 \pm 0.533708\%, \tag{32}$$
$$\text{MAE}_{\text{CPLA}} = 0.312034689 \pm 0.044063632 \text{ s}, \tag{33}$$
$$\text{MRE}_{\text{CPLA}} = 6.416740 \pm 0.454934\%, \tag{34}$$
$$\text{RMSE}_{\text{CPLA}} = 0.485851757 \pm 0.090484582 \text{ s}. \tag{35}$$

It improves CPLA-high MAE by 4.9% over the previous event-tubelet navigation JEPA mean (0.328 s), by 12.4% over navigation token JEPA (0.356 s), and by 32.9% over navigation token scratch (0.465 s).

9.2 Label Efficiency

At 10% labels, token JEPA improves diagnostic CPLA-high MAE by 65.4% over the matching scratch transformer. It also beats the full-label TinyCNN seed-7 baseline on diagnostic CPLA-high MAE (0.460 s versus 0.513 s).

9.3 Negative Architecture Ablations

The successful change was not simply larger scale or deeper supervision. The best local result came from matching the event structure with tubelet masking and preserving causal action/ego-motion conditioning.

9.4 Detection-Assisted Diagnostics

The causal bbox geometry reference is strong on CPLA-high bbox frames, showing that current bounding-box scale contains powerful TTC information. The frozen ROI latent prober improves strongly over its train-only ridge prior on validation, but it is worse than causal bbox geometry on CPLA-high and worse than the all-window JEPA relative error.

9.5 Predicted-Rollout Probers

The predicted-rollout prober tests the SkyJEPA idea more directly: probe frozen predicted future latents rather than only current context latents. Full-starter validation results are:

$$\text{MAE}_{\text{ROI latent}} = 0.226 \pm 0.015 \text{ s}, \tag{36}$$
$$\text{MRE}_{\text{ROI latent}} = 10.78 \pm 0.53\%, \tag{37}$$
$$\text{MAE}_{\text{ROI rollout}} = 0.226 \pm 0.013 \text{ s}, \tag{38}$$
$$\text{MRE}_{\text{ROI rollout}} = 11.15 \pm 0.79\%. \tag{39}$$

On the harder multi-validation split, ROI rollout is worse than ROI latent: weighted MAE 0.613 ± 0.017 s versus 0.528 ± 0.011 s. A dynamics feature mode with latent deltas and velocities reached only 0.248 s validation MAE on seed 7, worse than the flat seed-7 rollout prober at 0.210 s. These are negative results for the current flat-head prober.

10 Reproducibility

10.1 Environment

The final local verification used Windows, Python virtualenv `.venv`, PyTorch `2.11.0+cu128`, CUDA, an NVIDIA GeForce RTX 5070 Ti Laptop GPU, 32 GB RAM, and 32 CPU threads.

10.2 Final Commands

The repository state was verified with:

```
.\.venv\Scripts\python.exe -m ruff check .
$env:PYTHONPATH='src'
.\.venv\Scripts\python.exe -m pytest
```

After adding the official coverage gate, final verification was:

- `ruff check .`: all checks passed;
- `pytest`: 53 passed in 4.30 s.

10.3 Representative JEPA Command

The final JEPA pretraining family used the event-tubelet transformer with tubelet masking and a transformer dense predictor:

Table 2: Main full-starter all-window results. Lower MAE is better. Values are seconds unless noted. CPLA-high is diagnostic, not an official leaderboard test.

Method	Labels	Seeds	Validation MAE	Diagnostic CPLA-high MAE
Event-rate ridge	100%	det.	2.303	2.489
TinyCNN scratch	100%	7	0.549	0.513
Token transformer scratch	100%	7,13,21	0.702 ± 0.052	0.844 ± 0.008
Token JEPA + fine-tune	100%	7,13,21	0.358 ± 0.007	0.481 ± 0.042
Token transformer + navigation scratch	100%	7,13,21	0.440 ± 0.020	0.465 ± 0.021
Token JEPA + navigation fine-tune	100%	7,13,21	0.261 ± 0.021	0.356 ± 0.022
Event tubelet JEPA + navigation fine-tune	100%	7,13,21	0.243 ± 0.007	0.328 ± 0.030
Event tubelet JEPA + transformer predictor	100%	7,13,21	0.241 ± 0.004	0.351 ± 0.004
Event tubelet JEPA + tubelet mask + transformer predictor	100%	7,13,21	$\mathbf{0.231 \pm 0.018}$	$\mathbf{0.312 \pm 0.044}$

Table 3: Low-label evidence. Token JEPA gives the clearest benefit when TTC labels are scarce.

Method	Labels	Val. MAE	CPLA MAE
Token transformer scratch	5%	1.226 ± 0.031	1.382 ± 0.044
Token JEPA + fine-tune	5%	0.524 ± 0.047	0.636 ± 0.109
Token transformer scratch	10%	1.178 ± 0.056	1.327 ± 0.104
Token JEPA + fine-tune	10%	0.437 ± 0.039	0.460 ± 0.029

```
$env:PYTHONPATH='src'
.\.venv\Scripts\python.exe -m e_jepa_ttc pretrain jepa
  --cache <full-starter-nav-cache.npz>
  --output-dir <tubelet-mask-jepa-run-dir>
  --epochs 30
  --batch-size 64
  --learning-rate 5e-4
  --seed 7
  --device auto
  --model event-tubelet-transformer
  --pretrain-splits train
  --validation-splits validation
  --temporal-horizons-ms 20 60 100 240 500
  --max-target-slop-ms 10
  --mask-mode tubelet
  --dense-predictor transformer
```

11 Discussion

The evidence supports seven conclusions. First, JEPA pretraining helps event-based TTC in the local protocol. Second, the strongest benefits appear in label efficiency. Third, tokenization and dense prediction matter more than scaling a small encoder. Fourth, tubelet masking is the most successful V-JEPA-like modification tested locally. Fifth, causal navigation/action conditioning is useful, but not always positive in low-label settings. Sixth, detection-assisted ROI probers need more structure: flat latent probes and flat rollout probes are not enough. Seventh, official EvTTC SOTA remains unproven because only 37.5% of real-world official rows and 30.0% of complete Table V rows are locally covered, and the official baseline runtime protocol is not reproduced.

Table 4: SOTA-inspired ablations that did not become the selected local model.

Ablation	Result
Deep token JEPA	CPLA-high MAE 0.594 s
Deep layer-aware token JEPA	CPLA-high MAE 0.505 s
Large token transformer JEPA	CPLA-high MAE 0.529 s
All-token context loss, weight 0.25	Better car validation, worse pedestrian validation
All-token context loss, weight 0.05	Improved dev validation, did not beat full-starter validation
Tubelet mask plus all-token weight 0.05	Dev weighted MAE 0.649 s

The strongest defensible local claim is:

> On the local EvTTC full-starter all-window protocol, event-tubelet transformer JEPA with tubelet masking, dense transformer future-token prediction, and causal integrated-navigation conditioning achieves the best local result in this repository: 0.231 ± 0.018 s validation MAE and 0.312 ± 0.044 s diagnostic CPLA-high MAE over three fine-tuning seeds.

The strongest forbidden claim is:

> This is official EvTTC SOTA.

That claim is not supported by the current evidence.

Table 5: Detection-assisted bbox/ROI results. These are frame-label-only diagnostics and are not directly comparable with all-window metrics.

Method	Split	Labels	Predictions	MAE	Relative error
Causal bbox geometry	validation bbox frames	108	106	0.279	–
Causal bbox geometry	CPLA-high bbox frames	83	81	0.157	–
ROI event ridge	validation bbox frames	108	108	0.293	13.63%
ROI event ridge	CPLA-high bbox frames	83	83	0.829	47.12%
ROI latent prober	validation bbox frames	108	98	0.226 ± 0.015	$10.78 \pm 0.53\%$
ROI latent prober	CPLA-high bbox frames	83	73	0.423 ± 0.029	$23.41 \pm 2.23\%$

12 Limitations

The main limitations are:

- CPLA-high has been inspected and is diagnostic rather than pristine;
- official CCRs2, CCRm, and slider benchmark assets are missing locally;
- official STRTTC/CMax/ETTCM wrappers are not yet reproduced end-to-end;
- the event-tubelet backbone is small compared with V-JEPA-scale systems;
- deep self-supervision and all-token context loss were negative at the current data/model scale;
- no RGB, LiDAR, depth, or segmentation fusion is used during JEPA pretraining;
- no closed-loop planning or counterfactual action roll-out is evaluated;
- ROI rollout probing currently flattens horizons instead of enforcing TTC dynamics structurally.

13 Future Work

The next valid steps are not additional tuning on CPLA-high. They are:

1. download official CCRs2 and CCRm HDF5, TTC, and bbox/segmentation assets;
2. add the slider sequences if reproducing complete EvTTC Table V;
3. wrap or reimplement STRTTC, CMax, and ETTCM under deterministic CLIs;
4. evaluate every method on the same frames/events, sequence list, TTC alignment, metric, and runtime protocol;
5. build a horizon-structured TTC head over predicted latent rollouts;
6. tune VISReg/SIGReg-style regularization only on multi-domain validation;
7. run one final official benchmark once all assets and baselines are present.

14 Conclusion

E-JEPA-TTC demonstrates that dense event-token JEPA is a promising path for event-camera TTC estimation. The best local model combines event-tubelet tokenization, tubelet masking, dense transformer future-token prediction, EMA targets, and causal navigation/action conditioning. It reaches 0.312 ± 0.044 s diagnostic CPLA-high MAE and $6.42 \pm 0.45\%$ relative error, while low-label experiments show a 65.4% MAE improvement at 10% labels over the matching scratch token transformer. At the same time, the work is careful not to overclaim: the official EvTTC bbox/ROI benchmark is not yet covered locally, and current ROI probers are diagnostic rather than SOTA. The project is therefore best viewed as a reproducible, anti-leakage foundation for a future official EvTTC benchmark submission.

A Official Coverage Rows

B Main Artifact Pointers

Main all-window summaries live under `artifacts/metrics` with prefix *event tubelet, tubelet mask, transformer predictor, navigation, full-starter*. The key outputs are the validation and test `eval_full_protocol` summary JSON files.

ROI latent summaries use the prefix *ROI latent prober with event-tubelet transformer predictor and navigation*. ROI rollout summaries use the shorter `roi_rollout_prober_tubeletmask` family for both full-starter validation and dev multi-validation outputs.

C Repository Commit Context

The official coverage gate was added in commit:

Table 6: Official EvTTC bbox/ROI coverage encoded by the repository gate.

Row	Required assets complete	Local status	Notes
`CCRs-1-low-100%`	yes	present	HDF5, TTC, bbox/leftlabel, gt.hdf5
`CCRs-1-medium-100%`	yes	present	HDF5, TTC, bbox/leftlabel, gt.hdf5
`CCRs-1-high-100%`	yes	present	HDF5, TTC, bbox/leftlabel, gt.hdf5
`CCRs-2-low-100%`	no	missing	required for official real-world benchmark
`CCRs-2-medium-100%`	no	missing	required for official real-world benchmark
`CCRs-2-high-100%`	no	missing	required for official real-world benchmark
`CCRm-low-100%`	no	missing	required for official real-world benchmark
`CCRm-medium-100%`	no	missing	required for official real-world benchmark
`Slider-750`	no	missing	required for complete Table V reproduction
`Slider-1000`	no	missing	required for complete Table V reproduction

```
2f7103a eval(protocol): add official evttc coverage gate.
```

The present LaTeX article is derived from the validated markdown report and the committed protocol gate.

References

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